4.4 Detecting ranges with branches

Settings

Beta features

relational operator

A relational operator checks how one operand's value relates to ar
greater than.

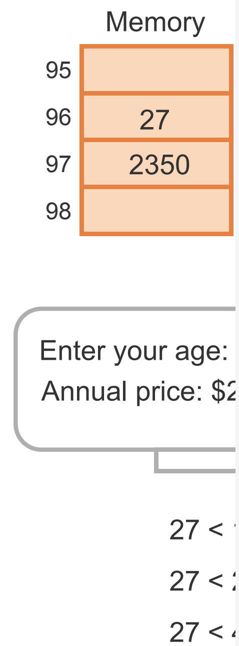
Table 4.4.1: Relational operators.

Relational operators	Description	Exampl
<	a **<** b means a is less than b.	x < 4 is x < 3 is
>	a **>** b means a is greater than b.	x > 2 is x > 3 is
<=	a **<=** b means a is less than or equal to b.	x <= 4 i x <= 3 i x <= 2 i
>=	a **>=** b means a is greater than or equal to b	x >= 2 i x >= 3 i x >= 4 i

PARTICIPATION ACTIVITY 4.4.3: Using the sequential nature of multi-branch if-else for ranges: Insurance pri

Start 2x speed

```
user_age = int(input("Enter your age: "))

if user_age < 16:             # Age 15 and under
  print("Too young.")
  insurance_price = 0

elif user_age < 25:           # Age 16 - 24
  insurance_price = 4800

elif user_age < 40:            # Age 25 - 39
  insurance_price = 2350

else:                         # Age 40 and up
  insurance_price = 2100

print(f"Annual price: ${insurance_price}")
```



Captions

1. The user enters 27 for their age, which is stored in memory as the variable user_age. The m user_age is less than 16, which is False.
2. The next branch in the multi-branch if-else checks if user_age is less than 25, which is False
3. The next branch checks if user_age is less than 40, which is True. The elif's statements exe insurance_price is set to 2350 in memory.

CHALLENGE ACTIVITY 4.4.1: Detect ranges using branches.

763650.1722066.qx3zqy7

Start

Type the program's output

```
x = int(input())

if x < 10:
    print("e")
else:
    print("n")

print("r")
```

Input

11

Output

```
n
r
```

1 2 3 4 5

Check Next

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CHALLENGE ACTIVITY | 4.4.2: If-else expression: Operator chaining.

Variable user_grade is read from input. Use operator chaining to complete the if-else expression a

- If the value of user_grade is between 9 and 12 (both inclusive), then "in high school" is outp
- Otherwise, "not in high school" is output.

Ex 1: If the input is 10, then the output is:

```
in high school
```

Ex 2: If the input is 8, then the output is:

```
not in high school
```

Note: 0 < x < 5 evaluates to true if x is between the ranges 0 and 5 (both non-inclusive).

Learn how our autograder works

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```
user_grade = int(input())

if """ Your solution goes here """:
    print("in high school")
else:
    print("not in high school")
```

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CHALLENGE ACTIVITY 4.4.3: Basic if-else expressions.

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Start

Complete the if-else statement to output "Under 70" if the value of user_num is under 70. Other

Ex: If the input is `68`, then the output is:

```
Under 70
```

```python
user_num = int(input())

if """ Your code goes here """:
    print("Under 70")
else:
    print("70 or more")
```

1

Check Next level

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CHALLENGE ACTIVITY 4.4.4: Working with branches.

763650.1722066.qx3zqy7

Start

Integer pet_count is read from input. If pet_count is greater than 15, then output "Enough pets"

Click here for examples

```
pet_count = int(input())

""" Your code goes here """

```

1 2 3

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CHALLENGE ACTIVITY 4.4.5: Working with ranges.

763650.1722066.qx3zqy7

Start

When the input variable outfit_count is:

- less than 6, output "Mid-sized suitcase".
- between 6 inclusive and 13 inclusive, output "Extra large suitcase".
- greater than 13, output "Need more than one suitcase".

Click here for examples

```
outfit_count = int(input())

""" Your code goes here """

```

1

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